STATISTICS: MEASURES OF CT AND DISPERSION

STATISTICS

DATA ORGANIZATION AND PRESENTATION

From the following data:

12	36	40	28	19	10	10	16	21	19	86	46
20	19	67	45	33	55	26	37	30	20	51	53
30	20	36	41	42	20	22	23	24	25	98	49
26	27	29	28	30	16	25	33	38	30	40	29
31	17	24	26	37	27	11	27	26	22	13	16
12	36	40	28	19	10	10	16	21	19	75	76
20	19	7	45	33	45	26	37	30	20	76	24
30	20	36	41	42	20	22	23	24	25	15	34
26	27	29	28	30	16	25	33	38	30	15	26
31	17	24	26	37	27	11	27	26	22	48	24
57	22	42	32	18	19	38	20	61	21	82	24
49	18	35	25	16	90	29	16	38	65	51	41
40	40	35	57	46	21	80	93	35	35	51	24
29	81	18	49	82	42	18	29	40	38	29	38
71	46	18	18	68	55	55	61	44	20	28	39
36	23	29	39	36	26	19	27	43	19	26	15
23	24	37	35	35	19	18	33				

Raw data

Required:

Types of classes
(i) Inclusive
(ii) Exclusive

1. Prepare a frequency distribution by the exclusive method (employ the Sturges Rule)

Range = $X_{max} - X_{min}$

Class width = h = Range / K

No. of classes = $K = 1 + 3.322 \log_{10} N$

2. Represent the data with a less than and a more than ogive; hence estimate the interquartile range from the graph

< ogive : plot more than c.freq against lower class intervals

> Ogive : plot < c.freq. against upper C.I.

3. Represent the data with a histogram and hence estimate the mode

vertical axis : freq Horizontal axis : classes

$\Rightarrow$ Modal value

4. Calculate the mean

$$\bar{x} = A + h\frac{\Sigma fd}{\Sigma f}$$

5. Calculate the median

$$M_d = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$

6. Calculate the mode

$$M_o = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

7. Calculate the Lower quartile

$$Q_1 = l + \frac{h}{f}\left(\frac{N}{4} - c\right)$$

8. Calculate the Upper quartile

$$Q_3 = l + \frac{h}{f}\left(\frac{3N}{4} - c\right)$$

9. Find range

$$\text{Range} = X_{max} - X_{min}$$

10. Determine the standard deviation

$$\delta = h\sqrt{\frac{\Sigma f d^2}{\Sigma f} - \left(\frac{\Sigma f d}{\Sigma f}\right)^2}$$

11. Calculate the coefficient of variation

$$C.V. = \frac{\delta}{\bar{x}} \times 100\,\%$$

1st Assgt: Determine/Describe/Explain the advantages and Limitations of using each one of the following measures in statistics:

(i) Mean (ii) Mode (iii) Median

Key Differences between Skewness and Kurtosis

This is the fundamental differences between skewness and kurtosis:

1- The characteristic of a frequency distribution that ascertains its symmetry about the mean is called skewness. On the other hand, Kurtosis means the relative pointedness of the standard bell curve, defined by the frequency distribution.
2- Skewness is a measure of the degree of lopsidedness in the frequency distribution. Conversely, kurtosis is a measure of degree of tailedness in the frequency distribution.
3- Skewness is an indicator of lack of symmetry, i.e. both left and right sides of the curve are unequal, with respect to the central point. As against this, kurtosis is a measure of data, that is either peaked or flat, with respect to the probability distribution.
4- Skewness shows how much and in which direction, the values deviate from the mean? In contrast, kurtosis explain how tall and sharp the central peak is.

12. Calculate the Karl Person's Coefficient of Skewness:

$$Sk_p = \frac{\text{Mean} - \text{Mode}}{\delta}, \qquad Sk_p > 0 \Rightarrow +\text{vely } Sk, \quad Sk_p < 0 \Rightarrow -\text{vely } Sk$$

$$Sk_p = \frac{3(\bar{X} - M_d)}{\delta}$$

13. Calculate Bowley's Measure of Skewness:

$$Sk_B = \frac{Q_3 - 2M_d + Q_1}{Q_3 - Q_1}$$

14. Calculate Kelly's Measure of Skewness:

$$Sk_K = \frac{P_{90} - 2P_{50} + P_{10}}{P_{90} - P_{10}}$$

15. Calculate Moment Measure of Skewness:

Coeff. of Sk based on moments is given by $\alpha_3 = \frac{\mu_3}{\delta^3} = \pm\sqrt{\beta_1} = \gamma_1$

where $\beta_1 = \frac{\mu_3^2}{\mu_2^3}$, $\mu_2 = \delta^2$

$$\mu_3 = \frac{h^3}{N} \Sigma f d^3, \text{ where } d = \frac{X-A}{h}$$

Moments about the mean:

$$\mu_r = \frac{h^r}{N} \Sigma f d^r \quad ; \; r = 0, 1, 2, \ldots$$

where $d = \frac{X-\bar{X}}{h}$

$$\mu_0 = 1 \qquad \mu_2 = \delta^2$$

$$\mu_1 = 0 \qquad \mu_3 = \frac{h^3}{N} \Sigma f d^3$$

$$\mu_4 = \frac{h^4}{N} \Sigma f d^4$$

Moments about an arbitrary point A:

$$\mu'_r = \frac{h^r}{N} \Sigma f d^r \quad ; \; r = 0, 1, 2, \ldots$$

where $d = \frac{X-A}{h}$

$$\mu'_0 = 1 \quad ; \; \mu'_1 = \frac{h}{N} \Sigma f d = \bar{X} - A$$

$$= \frac{h}{N} \Sigma f \left(\frac{X-A}{h}\right)$$

$$\mu'_2 = \frac{h^2}{N} \Sigma f d^2, \quad \mu'_3 = \frac{h^3}{N} \Sigma f d^3$$

$$\mu'_4 = \frac{h^4}{N} \Sigma f d^4$$

Conversion from Raw Moments into Central Moments:

$$\mu_0 = 1, \quad \mu_1 = 0$$

$$\mu_2 = \mu'_2 - (\mu'_1)^2$$

$$\mu_3 = \mu'_3 - 3\mu'_2\mu'_1 + 2\mu'^3_1$$

$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1'^4$$

16. Calculate Karl Person's Measure of Kurtosis:

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3$$

for a normal or Meso-Kurtic Curve

$\beta_2 = 3$ or $\gamma_2 = 0$

for a Lepto-Kurtic Curve $\beta_2 > 3$

or $\gamma_2 > 0$

for a Platy-Kurtic Curve $\beta_2 < 3$

or $\gamma_2 < 0$

2nd ASSGT

For the following set of data

Class	0-20	20-40	40-60	60-80	80-100
Frequency	13	25	27	19	16

Calculate:

1. Mean
2. Median
3. Mode
4. Standard deviation
5. Moment measure of Skewness
6. Moment measure of Kurtosis

ASSGT

The first four moments of a distribution about the value 4 of the variable are -1.5, 17, -30 and 108.

a) Find the moments about the mean, β_1 and β_2
b) Find also the moments about (i) the origin, and (ii) the point x=2

ASSGT

The yield of a particular crop on a farm is thought to depend principally on the amount of rainfall in the growing season. The values of the yield y, in tons per acre and the rainfall x. in centimeters, for seven successive years are given in the following table:

X	12	14	15	11	13	14	12
Y	6	8	8	5	7	9	6

(a) Find the equation of the least squares regression line of
(i) y on x
(ii) x on y
(b) Given that the rainfall in the growing season of a subsequent year was 14 cm, estimate the yield in that year

a) Given that the yield in a subsequent year was 8 tons per acre, estimate the rainfall in the growing season of that year

ASSGT

Ten competitors in a beauty contest are ranked by three judges in the following order:

1st Judge	1	6	5	10	3	2	4	9	7	7
2nd Judge	3	5	8	4	7	10	2	1	6	9
3nd Judge	6	4	9	8	1	2	3	10	5	7

Use the Spearman's rank correlation coefficient to determine which pair of judges has nearest approach to common taste in beauty.